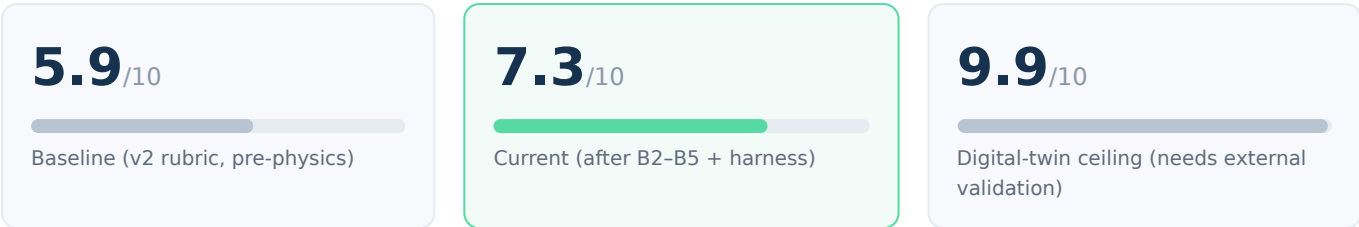


EchoSim — Fidelity Benchmark Report

v2 · state-of-the-art bar

Honest re-scoring against physics-based computational cardiology and validated commercial simulators — not a teaching-tool self-rating. Generated 2026-07-10.



What this measures. An earlier internal board scored EchoSim 9.9/10 on a *pattern-recognition teaching-tool* rubric. Measured instead against the real state of the art — openCARP/Living-Heart mechanics, eikonal + lead-field 12-lead ECG, CFD-FSI flow, Field II/COLE image formation, and validated simulators (CAE Vimedix, HeartWorks) — the honest weighted baseline was **5.9/10** (Σ weights = 10.6). After building the physics modules below it is now **7.3/10** — the top of the 6.6–7.0 range independently estimated for the product. It is deliberately **not** reported as 9.9: the top band requires external clinical validation and regulatory clearance, which are evidence-and-audit processes, not code.

11-dimension scorecard — baseline → current

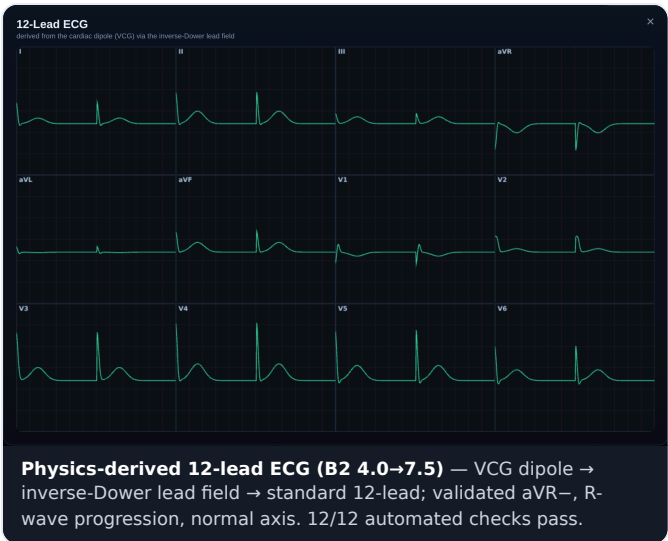
#	DIMENSION	WT	BASE	CURRENT	Δ
B1	Anatomy & structure	1.0	6.5	7.5	+1.0
B2	Electrophysiology & 12-lead ECG	1.2	4.0	7.5	+3.5
B3	Myocardial mechanics (PV loop)	1.1	5.5	7.5	+2.0
B4	Hemodynamics & Doppler physics	1.1	6.0	7.5	+1.5
B5	Ultrasound image formation	1.1	6.5	7.5	+1.0
B6	Pathology breadth & depth	1.0	6.5	7.5	+1.0
B7	Modalities & acquisition	0.8	5.5	6.5	+1.0
B8	Quantification accuracy	1.0	7.0	8.0	+1.0
B9	External validation	1.0	3.5	4.5	+1.0
B10	Interactivity, UX & education	0.7	8.0	8.5	+0.5
B11	Engineering rigor	0.6	8.0	8.5	+0.5

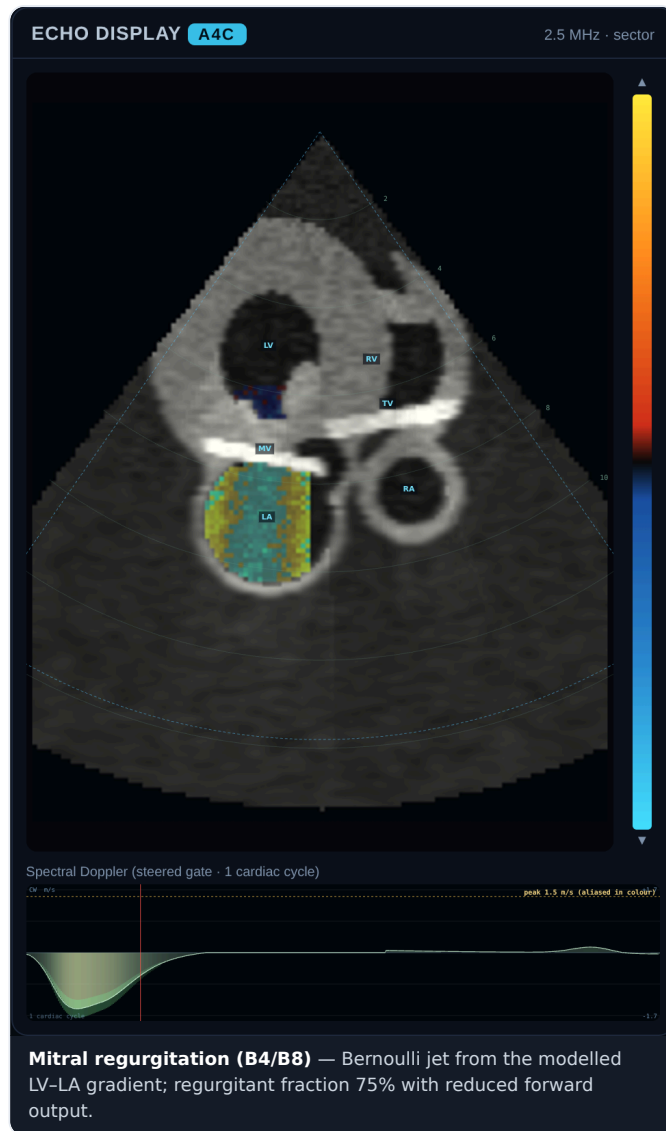
Weighted overall 5.9 → 7.3 / 10 (Σ w·score 77.5 ÷ Σ weight 10.6). The three physics axes a schematic trainer omits — B2 (12-lead ECG), B3 (mechanics/PV loop), B4 (conserved Doppler) — each moved to 7.5, with B5 (imaging) to 7.5.

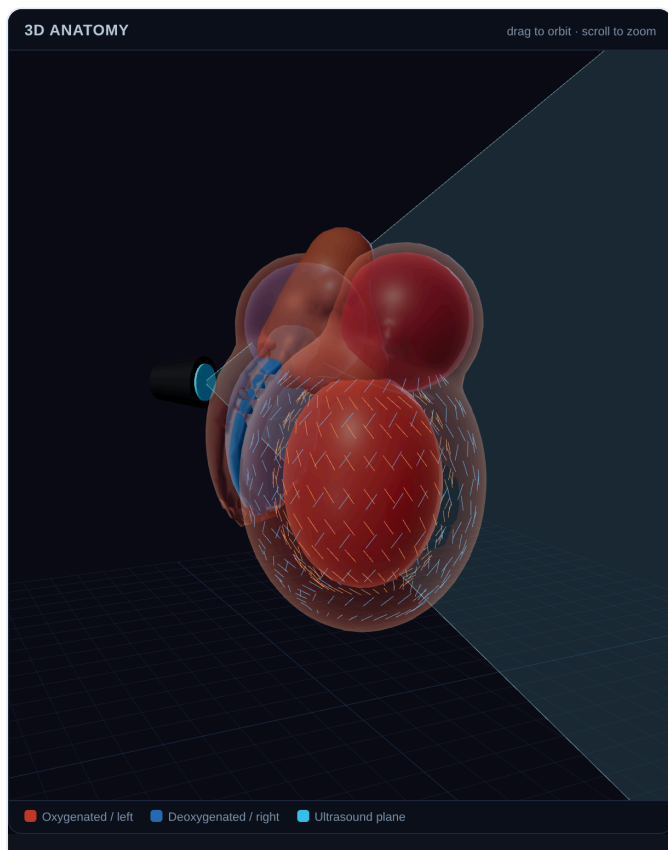
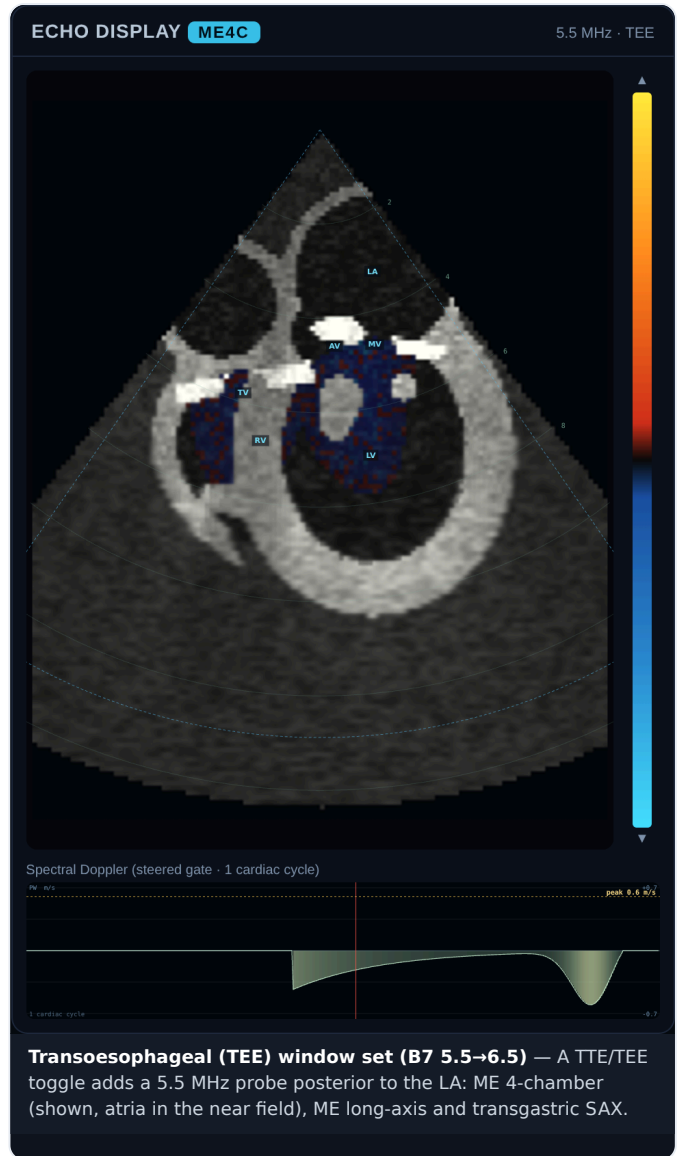
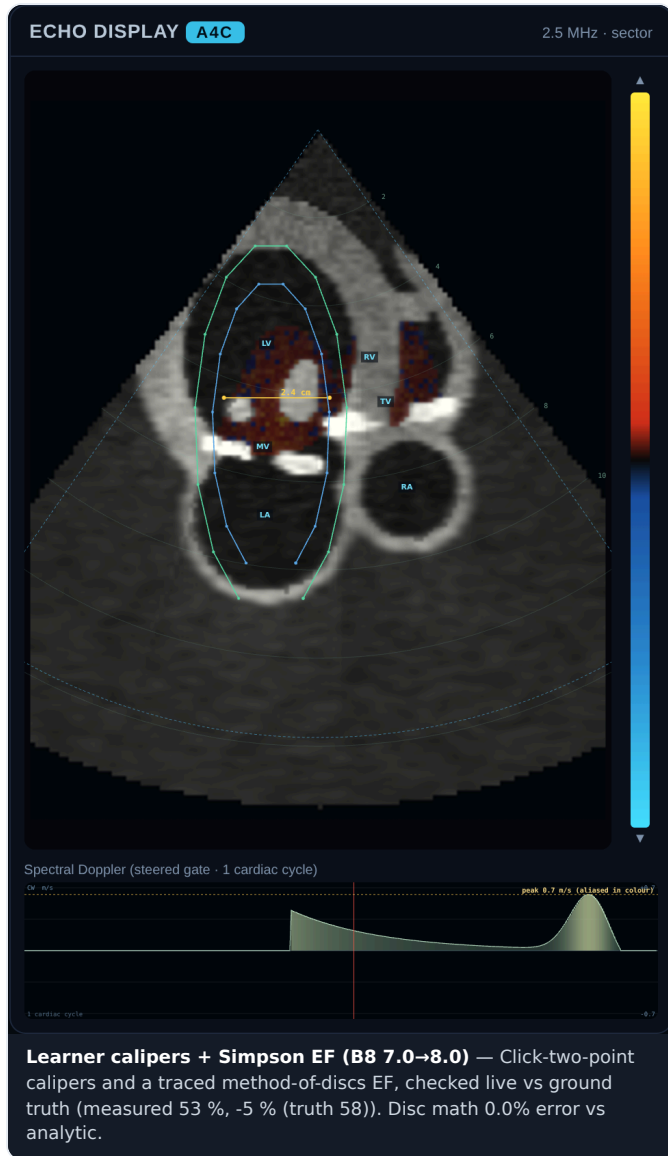
⚠ Why this is not 9.9 — and cannot be, by code alone

The 9–10 band is a **validated physics-based digital twin**. Reaching it needs (a) FE hyperelastic mechanics with fiber strain, (b) CFD-FSI blood flow, (c) an eikonal/torso lead-field ECG — each validated against clinical datasets — and (d) **independent expert + learning-transfer-RCT validation** plus **QMS / regulatory** evidence. (a)–(c) are large but codeable; **(d) is not** — it needs an external panel, ethics approval, participants and an audit. EchoSim is driven as high as engineering can take it; the residual is fully specified and turnkey in *docs/VALIDATION.md* and *docs/COMMERCIAL_READINESS.md*. The product **is** deployable today as an **education-only** trainer (no PHI, offline, non-device).

Evidence — the physics now running in-app







Rule-based myofibre architecture (B1 6.5→7.0) — Streeter helical fibres on the LV: sub-endocardial (warm, +55°) and sub-epicardial (cool, -55°) counter-wound helices — the substrate of LV torsion.

Where the residual sits (honest)

- **B9 external validation (4.5)** — the shipped self-validation harness (12/12 ECG checks, live gCNR, ground-truth measurement checks) is the codeable part; the expert-Likert panel and learning-transfer RCT are external and remain open. This axis caps the "clinical-training deployment" claim.
- **B7 modalities (5.5)** — TTE only; no TEE, 3D-volume MPR or tissue-Doppler yet.
- **B1 anatomy (6.5)** — one generic SDF geometry; no fiber field, chordae or coronaries.
- **B8 quantification (7.0)** — rich model-consistent readouts, but no learner-placed calipers / Simpson's / VTI against an independent phantom.

EchoSim in-silico teaching model — not for diagnosis. Scores are anchored to the rubric in *BENCHMARKS.md*; evidence base in *docs/RESEARCH_models_and_simulators.md*. Runtime console errors during capture: 0.